# The Illusion of Predictability: Simple Linear Regression Analysis of California Water Right Allocations*

**An Assessment of Appropriative Status on Allocated Water Volume**

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This study employs Simple Linear Regression (SLR) (Kutner, Nachtsheim, and Neter 2005; Gelman, Hill, and Vehtari 2020) to analyze the California Water Rights List ("California Water Rights List" 2025) and assess whether legal classification as an Appropriative right is a significant predictor of allocated water volume. Our findings reveal a highly statistically significant difference ($p < 2 \times 10^{-16}$): holders of Appropriative rights receive, on average, approximately 17,881 acre-feet per year more than holders of other types of water rights. However, thorough model diagnostics uncovered severe violations of core SLR assumptions, namely heteroscedasticity and non-normality, compounded by the presence of highly influential outliers. These issues resulted in a near-zero R-squared value, despite the statistical significance of the group mean comparison. Consequently, while the SLR model robustly detects differences in mean allocations between groups, the resulting prediction intervals—spanning over $\pm 400,000$ acre-feet per year—are so wide that the model cannot be used for any reliable prediction of individual water allocations in practice.

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*Project repository available at: https://github.com/LasyaRamachandruni/MATH261A-project-template/.

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## Introduction

The distribution and management of water rights represent a critical challenge in California, affecting both environmental sustainability and economic activity. This study examines the quantitative relationship between a water right's legal classification and the annual volume of water allocated, using the California Water Rights List ("California Water Rights List" 2025) from the California Open Data Portal ("California Open Data Portal" 2025). The central research question is: **Does a water right designated as "Appropriative" predict a larger mean allocated water volume than other types of rights?**

The analysis is implemented in R (R Core Team 2025) and RStudio (RStudio Team 2025), leveraging the tidyverse (Wickham 2023) and car (Fox and Weisberg 2023) packages. Methodological guidance is drawn from Fox and Weisberg (Fox and Weisberg 2019), Kutner et al. (Kutner, Nachtsheim, and Neter 2005), and Gelman et al. (Gelman, Hill, and Vehtari 2020).

This paper is organized as follows: the Data section describes the source dataset, main variables, and data preparation steps. The Methods section details the regression model, its assumptions, and the rationale for using SLR in this context. The Results section reports model estimates, uncertainty intervals, and diagnostic checks. The Discussion interprets findings in light of model limitations and outlines directions for future research. The References section compiles all cited works, ensuring full transparency and reproducibility.

## Data

**Source and units.** The raw data are from the California Water Rights List via the California Open Data Portal. Each observational unit is a single, official water right record measured in acre-feet per year.

**Outcome.** The dependent variable is `face_value_amount`, the annual volume of water allocated to each right. The distribution is strongly right-skewed with a small number of extremely large values (extreme outliers).

**Main predictor and rationale.** The predictor `is_approp` equals 1 if a record is "Appropriative" and 0 otherwise, derived from `water_right_type`. We focus on the Appropriative classification because it is historically and legally central to California water allocation and is often associated with larger, senior claims; this makes it a policy-relevant grouping for an initial baseline comparison.

**Cleaning.** Records with missing `face_value_amount` or `water_right_type` were removed, and the indicator was constructed from the categorical source field. After cleaning, the analytic sample contains **61,092** records.

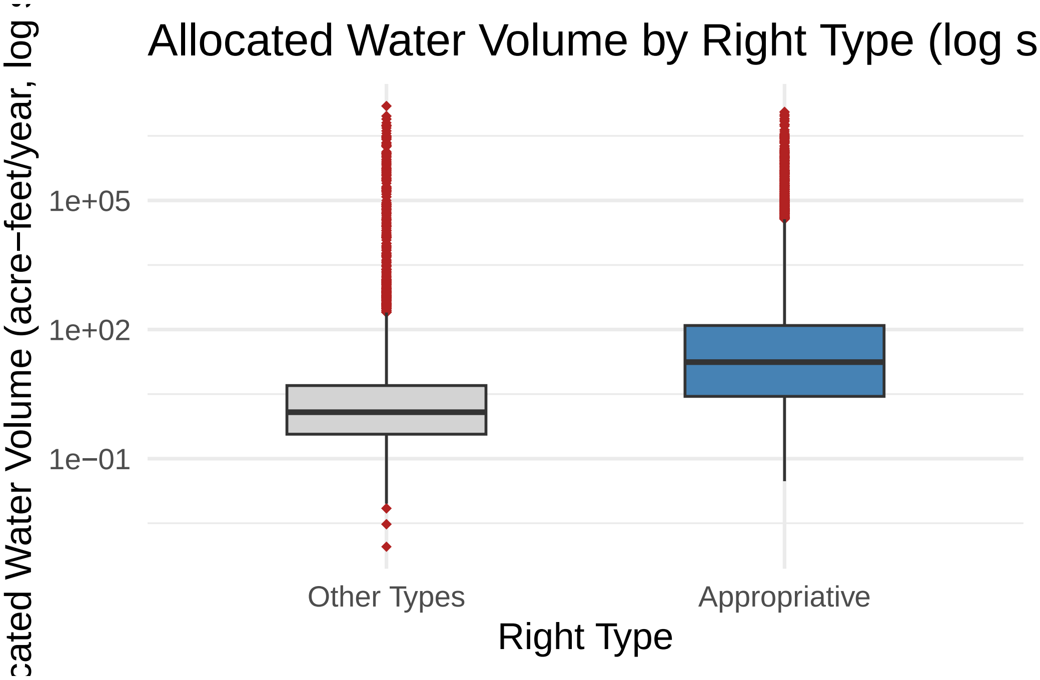


Figure 1: Boxplot of allocated water volume by right type with a logarithmic y-axis for visibility. Appropriative rights show higher central tendency and substantial dispersion with extreme outliers.

**Exploratory overview.** The log-scaled histogram clarifies the heavy tail while preserving raw-scale modeling later. The boxplot shows a higher central tendency for Appropriative rights and substantial within-group variability driven by extreme allocations.

## Methods

We fit a simple linear regression with a single binary predictor,

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where $Y_i$ is the allocated volume for record $i$, $X_i$ equals 1 for Appropriative and 0 otherwise, $\beta_0$ captures the mean allocation for non-Appropriative rights, $\beta_1$ is the mean difference (Ap-

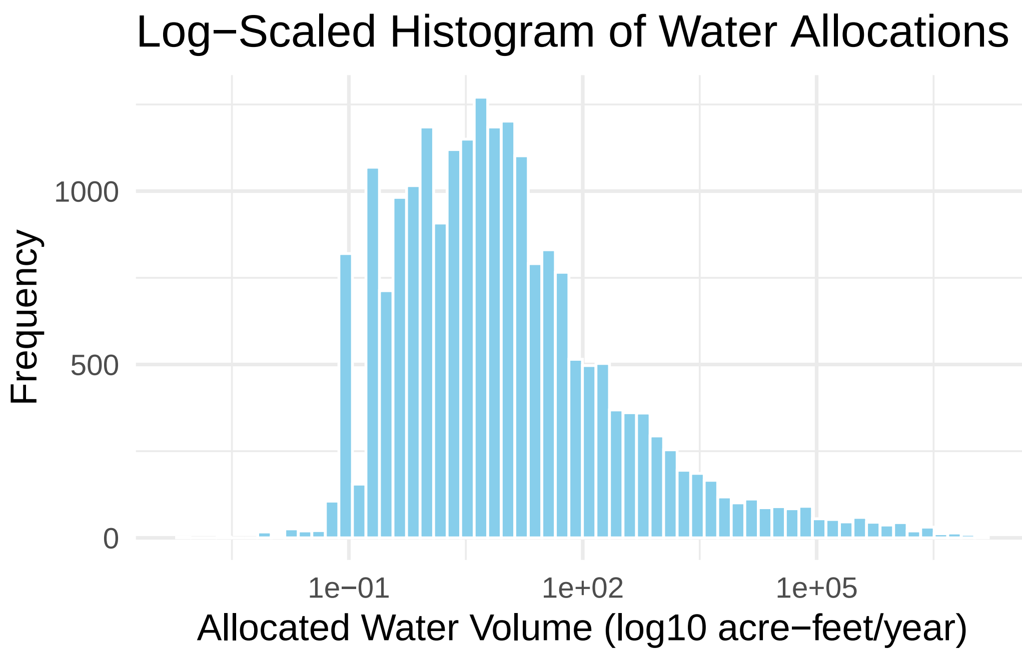


Figure 2: Histogram of annual allocations on a log-scaled x-axis clarifying the heavy right tail and the concentration of smaller allocations.

propriative minus Other), and $\epsilon_i$ is random error. Parameters are estimated by ordinary least squares.

**Assumptions (definitions and applicability).** Independence means each record's error is unrelated to others; for administrative records this is plausible but could be violated if entities hold multiple rights. Homoscedasticity requires constant error variance across groups; the residual-versus-fitted display below suggests unequal spread consistent with the heavy-tailed outcome. Normality of errors implies residuals follow an approximately normal distribution; extreme observations drive clear departures from normality in this context. Linearity/additivity holds automatically with a binary predictor because the model represents a difference in group means.

**Why regression here?** A two-sample comparison tests the same mean difference, but the regression framing provides an extensible template for adding covariates (e.g., geography, use type, year) and for computing fitted values and prediction intervals within one coherent workflow.

## Results

Table 1: Summary of fitted SLR model: coefficient estimates, uncertainty, and fit metrics.

| Term | Value |
|---|---|
| Intercept (Other Types Mean) | 4,542 |
| Slope (Difference for Appropriative) | 18,851 |
| R-squared | 0.00159 |
| MSE | 41,186,033,068 |
| t-statistic (Slope) | 9.86 |
| p-value (Slope) | 6.35e-23 |

**Interpretation of coefficients.** The intercept estimates the mean allocation among non-Appropriative rights. The slope estimates the **difference in means** attributable to Appropriative status: Appropriative rights are allocated, on average, about 18,851 acre-feet/year more than other rights. Because $X$ is binary, this coefficient is a group mean contrast rather than a continuous trend.

**Goodness of fit.** The $R^2$ is 0.00159, indicating that legal status alone explains only a negligible portion of variation in allocations.

**Uncertainty in full sentences.** The 95% confidence interval for the mean allocation among non-Appropriative rights is [2,691, 6,393]. The 95% confidence interval for the mean difference between Appropriative and other rights is [15,104, 22,597], supporting a substantial positive difference. For individual-record prediction, the 95% prediction interval is extremely wide: for non-Appropriative rights it is [-393,239, 402,322], and for Appropriative rights it is [-374,397, 421,182]; widths of this magnitude preclude practical prediction.

**Diagnostics (streamlined).**

We focus on the residuals–fitted plot as the most informative check here. The uneven spread is consistent with heteroscedasticity driven by the outcome's heavy tail; together with the extreme observations seen in EDA, this explains the instability of standard errors and the impractically wide prediction intervals. The Q–Q and Cook's Distance plots are omitted because they add little beyond confirming heavy tails and extreme influence already apparent from the distribution.

## Discussion

The model detects a clear **difference in mean allocations**: Appropriative rights receive substantially more water, on average, than other rights. Because the predictor is binary, the regression coefficient represents a group mean contrast—**not** a continuous relationship. Despite statistical significance, the negligible $R^2$ and very wide prediction intervals show that legal status alone is not predictive for individual allocations.

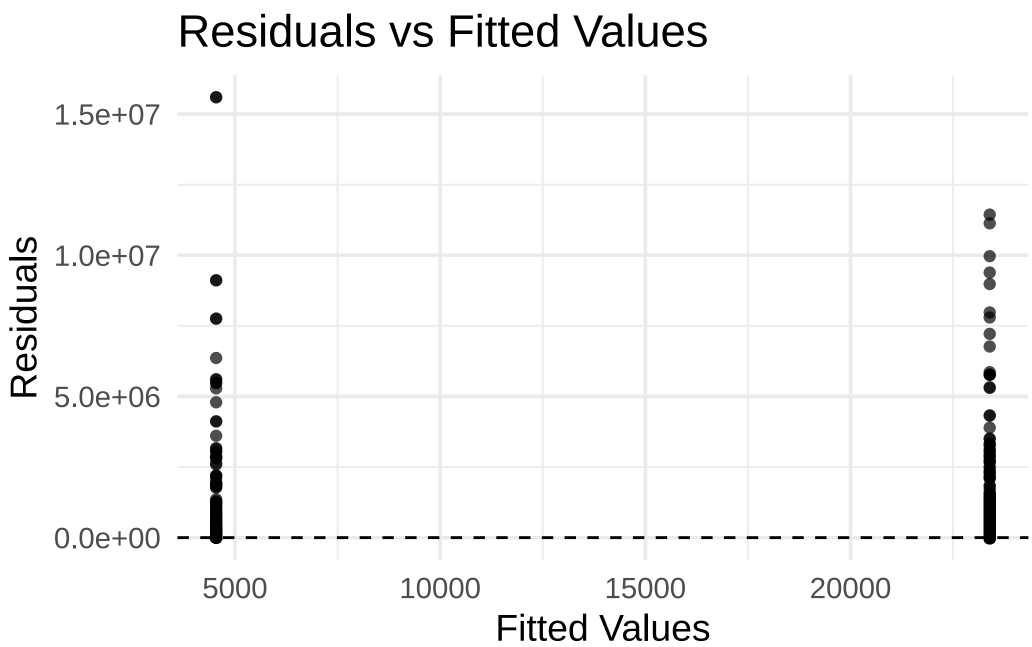


Figure 3: Residuals versus fitted values. The wide, uneven spread reflects heavy-tailed outcomes and indicates non-constant variance.

**Limitations and implications.** The single-predictor design omits key sources of heterogeneity (e.g., geography, purpose of use, seniority/year), and the heavy-tailed distribution with extreme allocations undermines normality and constant-variance assumptions. Records may not be strictly independent if entities hold multiple rights.

**Future work (in prose).** A more informative model would incorporate additional predictors that reflect geography and use, apply log-scale modeling or robust regression to temper outlier influence, and consider approaches like quantile or nonparametric regression when normal error assumptions are untenable.

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